

Operational Research

Queueing Simulations

The following exercises will require the use of Excel and Python. An example question and Excel spreadsheet is given on Course Resources. Please note that you need to use the same units, if you are using minutes, lambda and mu need to be in minutes.

Exercises

- 1) Implement a simple M/M/1 queue to model a single server in a bank.

Customers appear at a rate of 20 times per hour and the server can serve customers at a rate of 40 customers per hour. We could use $\lambda = 20$ and $\mu = 40$ if we stick to hours, however we will convert to minutes. For 20 customers to appear in an hour, we need 1/3 of a customer to appear every minute.

$$\lambda = \frac{20}{60} = \frac{1}{3}$$

$$\mu = \frac{40}{60} = \frac{2}{3}$$

You can use the following command in excel to generate the inter-arrival times and service times in minutes these have been set up for you.

$$-\frac{1}{\lambda} * LN(RAND())$$

$$-\frac{1}{\mu} * LN(RAND())$$

Theoretical Values

$$\text{Average Wait in Queue} = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$\text{Average Wait in System} = \frac{1}{\mu - \lambda}$$

Compare your theoretical values with the average simulation values. Plot two graphs to illustrate the convergence to the theoretical values. Also plot the time spent in the queue by each job and the time spent in the system. You can use the template for this one!

Decrease the number of arrivals (e.g. 10 arrive per hour, this would be a value of $\lambda = \frac{1}{6}$, i.e. $\frac{1}{6}$ of a customer arrives every minute or 1 customer arrives every 6 minutes). What do you notice about the system.

- 2) Create a simulation in which jobs arrive at a rate of 1 every 10 minutes (hint, work out λ by working out how many jobs arrive every minute). The jobs are processed by a machine which is modelled by a normal distribution. On average it takes the machine 5 minutes to process a job with a standard deviation of 1. What is (approximately) the **Average Time in Queue** and the **Average Time in System**? You can sample from a normal distribution in Excel using `NORM.INV(RAND(), mean, sd)`. By increasing the rate at which jobs arrive, approximate the value of λ at which point the system becomes unstable, i.e we get an infinite (unrecoverable) queue.
- 3) Extend your simulation from question 2) to simulate 3 machines in series. These machines should be normally distributed with mean and standard deviations, (5, 1), (2, 0.3) and (3, 0.5) respectively. Again, approximate the average time in queue and the time in the system. Is the system currently stable?
- 4) Repeat 1, 2 and 3 using Python. If you get stuck, you can ask AI. I want you to get used to how a simulation works and how you can analyse it by generating metrics like **Average Time In Queue**, etc. We will discuss this more when we do Discrete Event Simulation (DES).